

Search sheets: how to run the activity



The pages after this one are 24 **search sheets**, built from *The Tale of Peter Rabbit* by Beatrix Potter. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

Each sheet holds 46–47 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **thing he**, the boxed token is the previous token and *he* is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 12 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

1123 tokens, 388 unique · 2.25 bits/token · perplexity 4.8 · branching factor 2.32

Running a round

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair's next token.
4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

Worked example

Start anywhere. Pick any token pair — say this one — and write its previous token *and* its next token on the board.

Round 1. Call out the last token on the board (*he*). Another sheet has a match, so that person raises a hand and reads out *ate*.

Round 2. Call out the last token on the board (*ate*). Another sheet has a match, so that person raises a hand and reads out *some*.

Keep going for as long as you like. The text the room writes will sound like *The Tale of Peter Rabbit* without ever repeating it — that is the model generating, not the corpus being read out.

What to watch for

The hands are the probability distribution. Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Running dry. If no hands go up, you have reached a context that only ever occurred at the very end of the text. Start again from any token pair.

the token pair that answers

thing *he*

he **ate**

ate *some*

on the board

thing he

thing he ate

thing he ate some

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "thing", then the token pair **thing** **he** is a match and you answer **he**.

thing **he**

. **McGregor's**

but **don't**

was **locked**

gate **,**

, **underneath**

I **am**

in **that**

the **way**

small **for**

his **clothes**

root **of**

noise **of**

there **was**

and **peeped**

she **gave**

began **to**

, **and**

were **Flopsy**

, **he**

amongst **the**

after **a**

he **became**

excitement **,**

very **fast**

him **,**

a **blue**

up **the**

now **my**

upon **a**

at **last**

going **out**

rake **and**

. **and**

Peter **had**

walk **behind**

idea **which**

new **.**

run **into**

Rabbits **,**

to **the**

to **Peter**

running **as**

went **down**

the **tip**

. **rabbit**

garden **,**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "hole", then the token pair **hole** **and** is a match and you answer **and**.

cans .

hole **and**

friendly
sparrows

Peter **sneezed**

. **now**

up **for**

pie **by**

names **were**

big **fir**

of **breath**

at **bed**

loaf **of**

little **jacket**

his **jacket**

the **lane**

and **squeezed**

leaving **his**

. **but**

suddenly ,

very **quietly**

currant **buns**

were **four**

. **Peter**

sitting **in**

; **and**

she **wondered**

among **the**

after **him**

to **the**

, **went**

to **frighten**

the **other**

alive .

went **through**

gate !

, **if**

buns .

came **up**

was **a**

there ;

him **at**

in **a**

and **rushed**

, **who**

he **ate**

a **loaf**

get **into**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "last", then the token pair **last** **in** is a match and you answer **in**.

last **in**

big tears

the tool

the top

Peter **did**

and there

way to

up and

pond where

came to

in ,

been a

little Benjamin

him in

to squeeze

Cotton tail

he rushed

quietly off

and calling

very damp

. Peter

, and

. Also

back towards

find his

was no

there were

great
excitement

gate ,

after Peter

at him

clothes .

currant bushes

, whom

cooking ;

he was

she could

; he

if it

were good

a door

. he

to cry

of a

no time

some radishes

eyes .

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "flower", then the token pair **flower** pot is a match and you answer pot.

flower pot

the floor

he had

gate .

tea ;

in a

the fields

way straight

were overheard

was quite

he heard

him in

very naughty

and got

and started

little jacket

came out

, who

back was

. he

some lettuces

if he

sight of

staring at

of his

; he

head at

till he

Peter !

waving a

only shook

to go

soft sand

she had

big fir

. his

so much

after losing

Cotton tail

shut his

. McGregor

to wander

, but

a straight

no room

shook her

flopped down

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "calling", then the token pair **calling out** is a match and you answer out.

calling out

. McGregor

of running

; but

and beyond

if it

Bunny .

he might

little bunnies

Peter wriggled

gather
blackberries

might have

and beans

or down

a fortnight

umbrella ,

to her

wriggled out

safe at

family in

was Mr

to turn

care .

him to

, and

some friendly

! Mr

bed time

cabbages ,

scattered
underneath

frame ,

so that

. an

breath and

scr r

gate !

very still

way to

, Mopsy

the nice

old mouse

McGregor was

back to

in no

he had

. presently

she bought

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "across", then the token pair **across** the is a match and you answer the.

across the

close to

Peter began

! one

. Peter

tip of

bed ,

and out

he tried

without
speaking

bought a

or looked

some black

. McGregor's

to exert

so tired

lost ,

lippy lippy

filled his

cabbages ,

in her

to find

into a

old Mrs

, Mopsy

the bushes

way back

very big

back to

of shoes

outside the

a hoe

McGregor was

, scratch

French beans

accident there

was most

. now

and tried

little rabbit

; and

crow to

, but

still ,

he intended

peas and

fields or

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "made", then the token pair **made** **some** is a match and you answer **some**.

made **some**

go ,

look for

. **They**

gooseberry **net**

shed ,

doorstep ,

he **went**

to **rest**

put **him**

Peter **never**

and **shed**

baker's .

large **buttons**

plants .

, **and**

and **Peter**

he **ran**

; **and**

lippy **not**

was **the**

carrying **peas**

pair **of**

a **wall**

lost **in**

potatoes .

! **I**

. **but**

what **he**

the **window**

, **perhaps**

very ,

camomile **tea**

had **heard**

, **and**

into **a**

little **Rabbits**

sick ,

old **Mrs**

onions .

altogether **if**

some **parsley**

first **he**

of **Peter**

sneezed
Kertyschoo

frighten **the**

McGregor **filled**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "straight", then the token pair **straight** away is a match and you answer **away**.

straight away

the evening

large pea

sieve ,

shed ,

Once upon

out young

; but

. she

wheelbarrow
and

into mischief

done with

best to

a sieve

sat down

basket and

, and

carefully ,

bread and

to her

some gold

. she

put his

lost one

! Mr

had such

go into

Peter gave

and her

very well

he slipped

bunnies ,

first thing

the lane

spoonful to

, who

he found

McGregor came

sand bank

your Father

of a

, little

exert himself

and some

was very

hung up

fortnight !

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "end", then the token pair **end** of is a match and you answer **of**.

end of

least idea

three plants

straight walk

; but

radishes ;

he began

and more

second little

he could

some camomile

an old

found a

evening .

out of

presently Peter

had not

of a

sand on

water in

! Peter

through the

to be

go .

nothing
happened

Peter was

was tired

, Mopsy

, waving

sat very

cry .

. I

put in

, carrying

the garden

each .

the blackbirds

McGregor was

. he

and their

tail twitched

wheelbarrow ;

a beautiful

bread and

into the

shed big

thief !

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "and", then the token pair **and** **the** is a match and you answer **the**.

and **the**

had **so**

an **accident**

whom **should**

the **tool**

cousin ,

go **away**

he **flopped**

quite **close**

, **and**

out **of**

tired **of**

tail **had**

straight **across**

McGregor
caught

into **the**

ran **straight**

, **which**

shed ,

some **French**

and **milk**

presently ,

he **lost**

! **first**

sorry **to**

. **his**

losing **them**

round .

hidden
underneath

blackberries **for**

four **legs**

Peter .

just **in**

was **the**

. **his**

then **he**

, **leaving**

frightened ;

of **her**

water **cans**

knees **planting**

a **fat**

the **way**

white **cat**

r **ritch**

to **go**

; **he**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "of", then the token pair **of him** is a match and you answer **him**.

tail ,	of him	; he	presently ,
, and	was somewhere	four little	but now
saw was	round the	. rabbit	Peter .
turned towards	fast as	quite sure	! Peter
tired that	them ,	and went	, looking
buttons on	he had	climbed upon	McGregor .
into Mr	to the	had forgotten	. McGregor
the cabbages	and looking	, and	by some
then old	overheard by	foot upon	he began
ran on	the gate	you may	squeeze underneath
jumped out	out .	go into	a rake
ate some	blackberries :	may go	

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "milk", then the token pair **milk** and is a match and you answer **and**.

milk and

, and

he had

Peter ,

, but

my dears

taken at

got caught

was too

not answer

to pop

Mr .

of brown

then the

tree .

had not

he was

by the

Benjamin
Bunny

his water

much water

; and

and knees

their names

. a

ate some

Mrs .

them over

buttons ,

all round

but Mr

, and

and then

a window

puzzled .

cats from

the end

out just

the least

McGregor
hoeing

jacket .

. then

dreadfully
frightened

jumped into

quite new

ran after

fast ,

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "of", then the token pair **of a** is a match and you answer **a**.

Mr .	of a	, you	a wheelbarrow
Peter scuttered	with their	, he	could go
got home	the gate	was not	McGregor ,
he went	but presently	the garden	pop upon
, Mr	to a	tree .	, and
jacket and	their mother	then he	, he
his cousin	and five	Mopsy ,	scritch .
faster ,	had bread	said old	all over
Father had	; she	lane to	morning ,
he meet	out ,	jumped up	and the
heard the	somewhere in	beyond him	not had
Mrs .	by Mrs	started running	

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "with", then the token pair **with** sitting is a match and you answer **sitting**.

with sitting

got down

and he

heard about

his mother

and Cotton

of it

sure that

net ,

was busy

feeling rather

not very

took a

jacket with

. Mr

a pond

could not

Mr .

window ,

Mopsy ,

. it

he came

sparrows ,

Stop thief

behind some

, going

perhaps hidden

McGregor hung

out ,

Peter was

to him

but Peter

, he

had done

himself .

door in

don't get

a white

for Mr

towards Peter

lane ,

, so

Mrs .

the wood

then ,

the second

would have

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "most", then the token pair **most** **dreadfully** is a match and you answer **dreadfully**.

happened ,

most **dreadfully**

Mopsy ,

peeped over

rabbit to

don't go

. McGregor

himself up

his hands

floor of

a large

, quite

not
unfortunately

, quite

Also he

jacket and

the gate

Mr.

McGregor !

towards the

. McGregor

and Cottontail

Peter ,

got away

but his

as fast

had lost

from his

beans ;

window was

of the

answer .

was after

out over

can .

to gather

and pair

the little

he was

well during

, scratch

behind him

room for

with brass

a pie

for supper

lived with

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "wander", then the token pair **wander** about is a match and you answer **about**.

wander about

in great

as if

her umbrella

forgotten the

had not

can .

that Peter

tool shed

for some

, ran

Flopsy ,

and she

, who

who jumped

time .

twitched as

his work

sobs were

a flower

with fright

a scare

jacket behind

behind him

to the

the rabbit

be taken

he was

tears ;

but he

and then

flew to

, feeling

. and

Mr .

naughty ,

am going

fat little

Peter thought

say that

. he

rabbit one

was out

it were

the wood

not very

beans to

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "in", then the token pair **in** a is a match and you answer **a**.

in a

. Flopsy

which he

forhe

black currant

as nothing

the gate

caught sight

a can

it to

, but

, but

had an

brown bread

he came

and don't

brass buttons

away altogether

was so

the stone

, she

busy cooking

not care

Mr .

rabbit hole

Peter ,

a basket

her head

and he

. he

stone doorstep

Flopsy ,

Cottontail ,

with his

but Peter

fright ,

tool shed

who was

fir tree

never stopped

time there

his shoes

to put

. Peter

am sorry

that Peter

dose of

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "wood", then the token pair **wood to** is a match and you answer **to**.

wood to

, upsetting

home to

with a

but round

. McGregor

scare crow

away without

, and

caught by

, and

rabbit took

slipped underneath

. Peter

and implored

Peter was

to bed

rushed all

his way

was staring

as he

young cabbages

lettuces and

. Peter

Flopsy ,

which way

her the

tool shed

the wheelbarrow

the root

and ran

time ,

Mr .

should he

it had

a dose

not the

fir tree

that he

in it

shoes that

he jumped

who flew

meet but

for a

one of

locked ,

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "McGregor's", then the token pair **McGregor's** **garden** is a match and you answer **garden**.

Mr .

mischief .

McGregor's
garden

. **presently**

and **went**

rather **sick**

looked **behind**

: **but**

was **turned**

her **family**

but **it**

unfortunately
run

for **a**

, **and**

and **trembling**

nice **soft**

shoes **among**

rushed **into**

. **McGregor**

time .

looking **all**

his **eyes**

a **very**

think **he**

Peter **got**

that **I**

. **the**

the **noise**

They **lived**

top **of**

to **Mr**

it **was**

on **his**

turn **them**

one **table**

ritch ,

who **were**

he **went**

away **to**

in **the**

, **and**

wood **outside**

, **but**

the **garden**

bushes .

over **carefully**

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "upsetting", then the token pair **upsetting** **three** is a match and you answer **three**.

upsetting **three**

hoeing **onions**

bushes .

one **morning**

, **and**

Kertyschoo !

looking **under**

to **look**

pot .

shoes **for**

but **Peter**

hide **in**

and **climbed**

and **was**

, **he**

was **very**

beautiful **thing**

dears ,

her **mouth**

. **he**

bank ,

. **McGregor**

it **was**

underneath .

the **wood**

McGregor's
garden

, **Stop**

wood .

his **sobs**

a **gooseberry**

. **after**

he **had**

Mr .

: **your**

on **the**

wondered **what**

in **time**

over **the**

for **lost**

trembling **with**

that **she**

down **the**

the **gate**

time **he**

him **was**

scratch ,

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "and", then the token pair **and** jumped is a match and you answer jumped.

and jumped

her ;

it best

garden .

. it

damp with

pea in

cat was

tried to

. after

Peter ,

in a

that Peter

, for

, scritch

to him

did not

wall ;

a sand

Mr .

under each

scratch ,

the shoes

corner ,

his foot

he ate

more and

him .

became more

implored him

upon the

table spoonful

went back

the potatoes

on four

hands and

and made

over the

. but

down upon

squeezed under

was put

but she

mouth that

, but

underneath the

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "mother", then the token pair **mother** was is a match and you answer was.

mother was

underneath the

he got

stopped
running

. McGregor

asked her

Peter ;

her tail

fish ,

, and

Mr .

good little

gold fish

tried to

and shut

in the

, said

but he

down very

the baker's

his back

to his

. it

. McGregor

that can

upon Peter

a cucumber

it .

blue jacket

under the

was safe

went faster

shoe amongst

him till

garden :

about ,

, very

speaking to

have been

the corner

blackbirds .

more puzzled

on his

hoe scr

running or

over .

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "to", then the token pair **to say** is a match and you answer **say**.

to say

I think

supper .

five currant

thought it

was running

down the

during the

went to

but Flopsy

running after

, he

. Peter

he saw

garden ,

Peter asked

she only

upon a

him .

a time

now run

, along

in the

it was

gave a

legs and

work .

the big

underneath the

and then

Mr .

mother in

where Mr

the large

, and

such a

have got

. the

at some

along ,

off the

mouse was

about cats

began to

. Mr

his mother

When the token called out matches the boxed token of a token pair below, raise your hand, and read out that token pair's **next token** if you're picked. If more than one token pair matches, just go with the first one you spot. (A token is one word or one punctuation mark.) For example, if the token called out is "I", then the token pair **I** am is a match and you answer **am**.

I am

began to

the tool

to hide

his jacket

went back

run along

thing to

he was

in and

underneath a

cucumber
frame

down to

the first

and
blackberries

. McGregor

were alive

along a

but suddenly

, and

gave himself

garden ,

. Mr

parsley .

planting out

was on

a time

mother put

other shoe

she sat

too small

rest ;

intended to

, Cotton

at the

now and

up with

running in

going lippity

. then

him to

it would

Peter sat

upon the

after Peter

, as